Aaron Yang

Staff AI Engineer at Micron

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Online version: Notion resume

SUMMARY

Staff AI Engineer at Micron with 7 years of experience building and running a workflow and change management platform used by thousands of people every day across Micron's global fab, assembly and test sites. Design authority in an engineering org of about 20, well-versed in .NET Core, Angular, CI/CD, Kubernetes and Azure. Cares deeply about code quality, observability and developer experience, and lately brings AI into everyday engineering: AI code review backed by evals, agent-written tests and MCP servers.

WORK EXPERIENCE

Micron, Taichung, Taiwan — 2019-11 – Present

Staff AI Engineer (2026-10 –) · Staff Software Engineer (2025-10 – 2026-10) · Senior Software Engineer (2023-10 – 2025-10) · Software Engineer (2019-11 – 2023-10)

Core platform — used by thousands of people every day across global fab, assembly and test sites

- Built and run an internal workflow and change management platform integrated with manufacturing systems, so changes are validated against business plans, approved, and applied back automatically; brought four new manufacturing facilities onto it.

Technical leadership — design authority in an engineering org of about 20

- Design authority for the team's largest programs: led requirement workshops that turned business processes into IT requirements, then broke them into developer-ready work.
- Technical lead for a manufacturing setup automation program (about 94% less setup time); took over a stalled cross-team conversion program, redesigned it and handed delivery to another team (first wave live in 2026).
- Stood up a standalone instance for a fab divestiture and handed it over on the sale's schedule; the team was named IT Culture Champions (2021).

Performance and cost — slowest operations from minutes to seconds

- Removed N+1 and per-record database access in two phases — batched operations across the services, then bulk-write support in the shared platform library — taking the slowest operations from minutes to seconds (over 100x).
- Designed the move of inline images out of the database into object storage: database size down about 99%, database CPU from saturated to under 10%, memory per database server down 75%, weekly API outages to zero.
- Rebuilt the data-warehouse export as incremental jobs (data freshness from about 12 hours to about 5 minutes), and cut a cloud monitoring bill by about 87% by removing duplicated telemetry.

Cloud and reliability — on-call from 60+ to under 5 hours a week

- Migrated services from Windows Server IIS to Kubernetes with Jenkins, Octopus, Docker and Helm, and adopted Azure AD, Key Vault, Service Bus, Blob Storage, Cache for Redis and Application Insights.
- Replaced manual server and VM restarts with Kubernetes health checks, liveness probes, autoscaling and global load balancing: manual API restarts fell from 20+ a week to zero, and on-call effort from 60+ to under 5 hours a week.
- Built the platform's observability over six years: centralized logging, browser-to-database distributed tracing (W3C trace context), per-query database timing and alert rules, used to catch recurring faults before users report them.

Developer experience and AI-assisted development — test coverage from about 40% to 85%

- Migrated the team's repositories and CI to GitHub and GitHub Actions, which unlocked GitHub Copilot, and cut CI feedback time by about 40%.
- Turned the team's code-review judgement into repository rules for AI code review, A/B-tested with evals; raised test coverage from about 40% to 85% with agent-written tests under team rules; set up MCP servers and team Copilot skills.
- Talk: “Less is More — Designing Agent Context Files from Paper to Production” (2026).

5xRuby, Taipei, Taiwan — Full Stack Engineer (Contract) — 2017-08 – 2017-11

- Shipped new features for dappei.com with two senior engineers and a PM in 3 months, including Instagram and Facebook OAuth login and scheduled posting through their APIs.

SELECTED PROJECTS

- **Danmu Classroom** (2018, rewritten 2026): audience messages from phone or LINE fly across the speaker's screen; Cloudflare Workers, Durable Objects over WebSocket, and an Electron overlay.
- **Bisection Anything** (2026): rank anything by pairwise comparison, with single elimination or a full ranking.
- **Master's thesis: Dcard post analyzer** (2017 – 2019): crawler plus sentiment analysis, BERT, and a mention graph to help experts judge conversation health. Thesis
- **Chinese Segmentation as Service** (2019): jieba, SnowNLP, and pkuseg Chinese word segmentation served as an HTTP API with Flask.
- **Caten Church website** (2016 – 2019): Rails app for member accounts, event sign-ups, attendance reports, and permission management.

SKILLS

Backend: C#, .NET Core, Entity Framework Core, Ruby on Rails, Python, Flask, Node.js, RESTful APIs, message queues

Frontend: TypeScript, Angular, RxJS, Vue, Bootstrap, Sass, Electron

Testing and quality: xUnit, Moq, AutoFixture, FluentAssertions, Jest, ESLint, Prettier, SonarLint, dotnet format

DevOps and cloud: Jenkins, Octopus, Docker, Helm, Kubernetes, OpenShift, Azure (AKS, Key Vault, Service Bus, Blob Storage, Redis, Application Insights), Cloudflare Workers

Data: MSSQL, PostgreSQL, MySQL, MongoDB, Redis, RabbitMQ, Snowflake

AI-assisted development: GitHub Copilot, AI code review rules, evals, agent context files, MCP servers

EDUCATION

M.S. Information Management, National Sun Yat-sen University (2017 – 2019), GPA 3.8 / 4.3
Researched conversation health on social networks with text mining and NLP.

B.E. Automatic Control, Feng Chia University (2013 – 2017), GPA 3.6 / 4.0
Completed a second program in Computer Science.

LANGUAGES

Mandarin Chinese: Native

English: Professional working proficiency (TOEIC 750/990)